Tidal Hiccups and Oxic Sighs: Machine-Learning classification of oxygenation events in a transitioning coastal wetland

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Abstract

Coastal wetlands are converting former freshwater floodplains into tidally influenced marsh, reshaping the subsurface dissolved-oxygen (DO) regime that governs biogeochemical function. At Beaver Creek, Washington, a near-anoxic floodplain well (DO levels below detection ∼92% of the time), punctuated by episodic oxygenation events. These events fall into two physically distinct classes: abrupt, asymmetric "hot moments" driven by tidal/saline intrusion, and slower, symmetric "oxic pulses" driven by freshwater rise from precipitation/flooding. We formalize separating these regimes as a supervised classification problem using six years of co-located 5-minute DO, hydrology (water level, salinity, temperature), and weather measurements.

We compare two modeling tracks. The first uses tabular classification models on domain-engineered event-shape features encoding morphology of the DO vs time data, change in salinity and water-level, antecedent-precipitation, etc. The second directly uses raw multivariate time-series data either as inputs to a Convolutional Neural Network (CNN), or through automatic encoding into $\mathcal{O}(10^4)$ features capturing correlations over various time scales, fed into tabular models. Classification performance is reported through metrics like Confusion Matrix, F1-score, Balanced Accuracy, etc.

Furthermore, through feature importance attribution, we have are isolating and identifying the various drivers for these DO events.